Software Requirements Specification (SRS)

Project Title: Salon Management Portal

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) outlines the comprehensive architectural, functional, and interface design requirements for the frontend client of the Salon Management Portal. This system operates as a single-page frontend application designed to interface with backend APIs to eliminate manual, paper-based records, eliminate scheduling conflicts, optimize staff tracking, and increase overall client retention. This engineering specification serves as the complete development guide for establishing client state synchronization, role-based UI access, dynamic multi-step booking forms, and secure asynchronous token persistence.

1.2 Scope

The functional core of this application focuses strictly on advanced frontend presentation development using ReactJS. The system establishes an interactive operational portal handling four distinct system client workflows:

- Customer Interface: Registration, profile adjustments, service browsing, product orders, and multi-step booking pipelines.
- Barber/Stylist Panel: Direct assignment monitoring, appointment state updates, working availability management, and commission earning logs.
- Receptionist Matrix: Multi-stylist master reservation management, client profile verification lookup, and barber scheduling allocation controls.
- Administrator Core Dashboard: Employee CRUD management, pricing controls, inventory stock alert configurations, commission tier updates, and visual data chart metrics.

1.3 Definitions

Term	Meaning
JWT	JSON Web Token; an encoded token structure used to maintain client identity securely inside HTTP authorization headers.
Route Protection	Client-side routing guards that evaluate local state user flags before permitting interface mounting.
Redux Toolkit	An advanced state configuration utility providing centralized slice storage to maintain intricate e-commerce carts and session variables.

Term	Meaning
CRUD	Create, Read, Update, Delete; standard operational data entry views executed using modal forms and data grids.
Multi-Step Wizard	A linear multi-pane layout component managing sequential workflow steps with isolated step validation tracking.

1.4 References

- Provided Frontend System Track Specification File: AFWT_Project_Description.docx.
- Core Component Library Framework: Material UI Layout Design Metrics.

2. Overall Description

2.1 Product Perspective

The Salon Management Portal behaves as a decoupled, standalone user interface engine. It manages state transformations locally via Redux Toolkit and shifts data processes externally via standardized JSON REST endpoint integrations.

Movie Information Web Application

|

├— Frontend Layer (React.js)

| ├— Material UI Components

| ├— React Router

| ├— React Hook Form

| └— Redux Toolkit

|

├— Storage Layer

| ├— Local Storage

| └— JWT Token Storage

|

├— Communication Layer

| └— Axios Client

|

├— API Layer

| ├— TMDB API

| ├── Search API

| ├── Movie Details API

| └── Trending Movies API

|

└── User Interface

├── Home Page

├── Search Page

├── Movie Details Page

├── Trending Movies

└── Top Rated Movies

Technical Specifications Stack:

- View Layer Foundation: ReactJS leveraging component decomposition design.
- Routing Guard Infrastructure: React Router protecting access tokens across paths.
- Centralized State Layer: Redux Toolkit keeping e-commerce cart states synchronized across views.
- Form Automation Layout: React Hook Form managing data input validation.
- Data Visualization Library: Recharts or Chart.js mapping real-time operational records.
- Static Deployment Server: Optimized and hosted on Vercel.

2.2 User Roles & Matrix Permissions

The portal manages user permissions across four distinct operational roles:

Role	**Permitted Presentation Access Scope**
Customer	Account setup, service catalog navigation, booking calendar wizard, retail store checkout, history data tables.
Barber / Stylist	Assignment sheet views, calendar status controls, earnings overview charts, schedule blocker forms.
Receptionist	Full master scheduling grids, customer lookup views, stylist assignment management.

Role	Permitted Presentation Access Scope
Administrator	Employee metrics configurations, service price adjustment fields, inventory supplier trackers, wage tables, business analytics dashboards.

2.3 Operating Environment

- Platform Scalability: Responsive layouts providing visual consistency across smartphones, tablets, laptops, and desktop workstations.
- Execution Compatibility: Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge.
- Connectivity Constraints: Requires persistent client internet connections to preserve data communication with the provided APIs.

3. Client-Side State Schema Definitions (Data Shapes)

Students must strictly structure their global state storage states, data tables, and forms to match the following client-side object blueprints:

3.1 Central Authentication & User Profile Slice

JavaScript

```
{
  auth: {
    token: "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", // Kept in persistent local storage
    isAuthenticated: true,[cite: 3]
    user: {[cite: 3]
      id: "usr_99823",[cite: 3]
      name: "Alex Mercer",[cite: 3]
      email: "alex@example.com",[cite: 3]
      role: "Customer" // Options: Customer | Barber | Receptionist | Admin
    }
  }
}
```

3.2 Dynamic Appointment Payload Booking Layout

JavaScript

```
{
  bookingWizard: {[cite: 3]
    selectedService: { id: "srv_102", name: "Beard Styling", price: 25.00, duration: 30 },[cite: 3]
```

```
  selectedBarber: { id: "emp_441", name: "Marcus Vance" },[cite: 3]
  selectedDate: "2026-07-15",[cite: 3]
  selectedTimeSlot: "14:30",[cite: 3]
  validationState: { isCurrentStepValid: true }[cite: 3]
 }
}
```

3.3 Shopping Cart State Shape

JavaScript

```
{
 cart: {[cite: 3]
  items: [[cite: 3]
   { id: "prod_051", name: "Premium Hair Wax", price: 18.50, quantity: 2 }[cite: 3]
  ],[cite: 3]
  totalQuantity: 2,[cite: 3]
  totalAmount: 37.00[cite: 3]
 }
}
```

4. Functional Modules & UI Requirements

4.1 Module 1: Authentication

- Features: Secure login fields, user onboarding registration flows, password reset assistance forms, and internal client role management checking.
- Frontend Design Layer:
    - Integrate strict component validations using React Hook Form to intercept empty submission payloads.
    - Establish structural Route Protection setups blocking unauthenticated routing requests.
    - Cache returning session keys inside local web storage containers to maintain session continuity during browser refreshes.

4.2 Module 2: Customer Dashboard

- Features: Summary layout showing upcoming salon visits, history datatables, favored treatments, and past transactional purchase histories.
- Frontend Design Layer:

- Implement stylized card widgets that highlight urgent calendar items prominently.
- Mount adaptive data grid elements supporting filter controls to allow deep historical log analysis**.**

4.3 Module 3: Service Management

- Features: Multi-category catalogs sorting items, duration calculations, and clear price information.
- Frontend Design Layer:
  - Group listings by categories like Haircut, Beard Styling, Hair Coloring, and Facial.
  - Bind client-side filter bars to allow users to screen treatments based on price caps or duration lengths.

4.4 Module 4: Appointment Booking

- Features: Sequential step wizards coordinating treatment choices, stylist selections, calendar evaluations, and slot reservations.
- Frontend Design Layer:
  - Implement custom interactive calendar grids indicating availability states clearly to prevent scheduling overlaps.
  - Construct a multi-step dynamic entry wizard with runtime step verification rules to enforce data integrity.

4.5 Module 5: Appointment Management

- Features: Complete appointment lists, cancellation actions, date adjustments, and manual status updates.
- Frontend Design Layer:
  - Implement color-coded status badges reflecting current scheduling configurations: Pending, Confirmed, In Progress, Completed, or Cancelled.

4.6 Module 6: Staff Management

- Features: Profile setups, talent tag filters, active shifts configuration, and exception time-off tracking tools.
- Frontend Design Layer:
  - Build interactive tabular administration interfaces supporting quick insert, read, update, and delete modal components.

4.7 Module 7: Wage Management

- Features: Compensation trackers calculating earnings, dynamic commission payouts calculation, and overall month-to-date summary tables.
- Frontend Design Layer:

- Design summary metric tiles presenting operational counters: Total Services Completed, Total Earnings Accumulated, and Active Commission Percentages.

4.8 Module 8: Inventory Management

- Features: Stock ledger logs tracking products, warning triggers highlighting low balances, and supplier reference modules.
- Frontend Design Layer:
  - Program programmatic threshold alarms that color-code item rows when core product assets (e.g., Shampoo, Hair Wax, Hair Color, Face Cream) fall beneath target safe stock volumes.

4.9 Module 9: Product Store

- Features: Retail catalog showcases, search controls, classification filtering widgets, shopping cart counters, and payment form outlines.
- Frontend Design Layer:
  - Configure global slice structures to preserve item additions, quantity updates, and cart subtotal calculations throughout browsing workflows.

4.10 Module 10: Analytics Dashboard

- Features: Corporate financial analysis charts, user base growth charts, service popularity matrices, and personnel performance trackers.
- Frontend Design Layer:
  - Utilize Recharts or Chart.js packages to assemble interactive visual representations of core company metrics.

5. Non-Functional Requirements

- Performance Metrics: Optimizing client assets and using structural code-splitting techniques to achieve page load metrics under 3 seconds.
- Security Frameworks: Application routes must check active authorization tokens before allowing view components to mount, handling authorization parameters safely within layout trees.
- Usability Criteria: Interface layouts must scale using responsive breakpoints to ensure consistent accessibility on handheld form factors and desktop workstations.
- Scalability Standards: Separate views into modular, reusable presentation blocks to simplify structural scaling and ease upcoming application expansions.

6. 12-Week Development Plan

The implementation schedule strictly follows the exact structural week titles specified within your educational track blueprint:

Week 1 – Requirement Analysis & UI Planning (22/06/2026 to 28/06/2026)

- **Tasks: Thorough evaluation of multi-role user requirements, constructing use-case interactions, mapping structural application wireframes, and setting up the folder structure layout.**
- **Student Project Tasks:**
    - **Map out the complete cross-role application permission table, detailing the route access configurations for all four user roles.**
    - **Sketch low-fidelity structural design wireframes for the user booking steps, e-commerce storefront layout, and the administration analytical board.**
- **Deliverable Output: Approved comprehensive requirement outlines and interface wireframe layouts.**

Week 2 – Environment Setup & Routing (29/06/2026 to 05/07/2026)

- **Tasks: Initializing the React framework base workspace, creating system path navigations, designing overall frame shells, and building global header links.**
- **Student Project Tasks:**
    - **Initialize the application codebase using efficient contemporary developer toolkits bundled alongside the necessary Material UI extensions.**
    - **Configure browser navigation rules using React Router components to establish separate access boundaries for public, client, and admin view tracks.**
- **Deliverable Output: Functional skeleton layout code integrating responsive application navigation routing setups.**

Week 3 – Authentication Module (06/07/2026 to 12/07/2026)

- **Tasks: Programming login layouts, account registration wizards, password reset workflows, and implementing client-side routing guards.**
- **Student Project Tasks:**
    - **Assemble validation-backed authentication components using input verification logic via React Hook Form hooks.**
    - **Build high-order routing guard wrappers that evaluate active authorization tokens before granting entry to specific dashboard segments.**
- **Deliverable Output: Complete working session authorization system enforcing proper route protection boundaries.**

Phase – 1 Evaluation (13/07/2026 to 17/07/2026)

Week 4 – Service Management Module (13/07/2026 to 19/07/2026)

- **Tasks: Building product listings dashboards, layout panels showing deep service specs, and constructing text search inputs.**
- **Student Project Tasks:**

  - **Program the main service card component array to neatly render service metadata across category folders (e.g., Haircut, Beard Styling, Coloring).**
  - **Implement client-side filter hooks that allow users to screen treatments based on price caps or duration settings.**
- **Deliverable Output: An interactive service discovery portal featuring accurate data search sorting tools.**

Week 5 – Appointment Booking Module (20/07/2026 to 26/07/2026)

- **Tasks: Engineering step-by-step booking assistants, date selection calendar tables, and time slot allocation frameworks.**
- **Student Project Tasks:**
  - **Build out a fluid, multi-step booking form wizard tracking step progression cleanly within local component states.**
  - **Create interactive schedule grids that dynamically grey out unavailable slots based on live stylist availability data.**
- **Deliverable Output: Fully validated booking wizard components capturing structural reservation choices.**

Week 6 – Customer Dashboard (27/07/2026 to 2/08/2026)

- **Tasks: Structuring historical summary logs tables, tracking upcoming visit lists, and crafting personal profile configuration inputs.**
- **Student Project Tasks:**
  - **Design specialized, card-based metric blocks that display real-time statuses for scheduled services.**
  - **Construct user-profile forms that hook directly into API endpoints to allow seamless account update operations.**
- **Deliverable Output: Functional user dashboard hubs displaying personalized booking records.**

Week 7 – Staff Management (03/08/2026 to 09/08/2026)

- **Tasks: Writing personnel directory tabular sheets, detailed biography sheets, and managing shift rosters tables.**
- **Student Project Tasks:**
  - **Construct administrative personnel rows displaying employee skill sets, active work hours, and scheduled time-off records.**
  - **Implement filtering panels that allow receptionists to scan and identify available stylists by specific skill tags.**
- **Deliverable Output: Completed personnel monitoring screen layouts handling core scheduling configurations.**

Phase – 2 Evaluation (10/08/2026 to 14/08/2026)

Week 8 – Appointment Administration (10/08/2026 to 16/08/2026)

- **Tasks: Organizing unified reservation summary listings tables, dropdown controls updating booking states, and staff assignment panels.**
- **Student Project Tasks:**
    - **Create a comprehensive reservation master grid for receptionists to review, filter, and track upcoming appointments.**
    - **Integrate quick state adjustment menus that pass status updates (e.g., Pending, Confirmed, In Progress, Completed, Cancelled) down to the API layer.**
- **Deliverable Output: Operational scheduling master control panel managing salon workflow configurations.**

Week 9 – Inventory Management (17/08/2026 to 23/08/2026)

- **Tasks: Setting up inventory summary matrix displays, product stock volume balance controls, and supplier registry listings rows.**
- **Student Project Tasks:**
    - **Build an administrative product ledger tracker showing accurate quantities on hand for products like Shampoo, Hair Wax, and Hair Color.**
    - **Apply warning highlights that flag inventory lines in red whenever stock levels fall below specified low-stock thresholds.**
- **Deliverable Output: Completed stock control portal featuring dynamic low-balance threshold warning triggers.**

Week 10 – Product Store (24/08/2026 to 30/08/2026)

- **Tasks: Arranging product store card displays, configuring persistent shopping cart drawers, and designing payment checkout forms.**
- **Student Project Tasks:**
    - **Set up global Redux Toolkit store slices to accurately handle real-time product item additions, quantity modifications, and item removal actions.**
    - **Program clean checkout summary pages that compute subtotals, item taxes, and total transaction amounts automatically.**
- **Deliverable Output: Complete, functional retail e-commerce store component module managing active cart states.**

Week 11 – Analytics Dashboard (31/08/2026 to 13/09/2026)

- **Tasks: Embedding business total revenue charts, user population metrics, treatment trend lines, and employee performance visual elements.**
- **Student Project Tasks:**

  - **Integrate Recharts area or line graph panels to display sales over time and visualize customer conversion records.**
  - **Construct multi-axis bar charts to evaluate employee performance by comparing completed appointments and commission percentages.**
- **Deliverable Output: Fully integrated business intelligence dashboard mapping real-time operational data.**

Week 12 – Testing, Optimization & Deployment (14/09/2026 to 20/09/2026)

- **Tasks: Executing software bug repairs, testing layout behaviors across mobile viewports, streamlining component rendering performance, and launching production site builds.**
- **Student Project Tasks:**
  - **Analyze rendering cycles and apply optimization patterns to trim overall bundle file sizes and secure sub-3-second load speeds.**
  - **Deploy the production compilation bundle out onto live Vercel servers, confirming that route protections and API communication keys remain functional.**
- **Deliverable Output: Complete live deployed portal instance, architectural project documentation, and final system demonstration slides.**

Phase – 3 Evaluation (21/09/2026 to 03/10/2026)

7. Evaluation Rubric Matrix

Student project portfolios will be strictly evaluated based on the following 100-mark distribution:

Assessment Component Criterion	**Weightage Maximum**	**Focus Verification Area**
UI Design & Layout Aesthetics	10 Marks	Consistent use of design systems, professional layout spacing, clean typography choices, and implementation of theme styling rules.
React Component Decomposition	15 Marks	Effective separation of components into reusable blocks, logical use of property inputs, and proper management of local states.
Role-Based Routing & Security	10 Marks	Reliable path protection configurations, accurate JWT token verification handling, and proper role-based interface access control.

Assessment Component Criterion	**Weightage Maximum**	**Focus Verification Area**
API Asynchronous Integration	20 Marks	Professional setup of Axios configurations, proper handling of asynchronous lifecycles, and reliable handling of error boundary conditions.
Global State Orchestration	15 Marks	Clean utilization of Redux Toolkit slices to maintain intricate application configurations and handle cart state synchronization.
Interactive Booking System	10 Marks	Valid step progression tracking, reliable prevention of scheduling slot overlaps, and clean design of calendar components.
Responsive Screen Adaptations	10 Marks	Smooth layout adjustments across simulated mobile, tablet, and widescreen monitor viewports using fluid design properties.
Code Hygiene & Organization	5 Marks	Clear folder categorization, descriptive and accurate naming conventions, and proper separation of layout logic from data processes.
Vercel Cloud System Deployment	5 Marks	A fully compiled, functional production application running live on remote Vercel cloud web servers.
TOTAL SCORING CAPACITY	100 Marks	Note: Extra marks credit targets may apply upon flawless completion of core objectives.

8. Expected Learning Outcomes

Upon successful integration and final presentation assessment of this portal project, students will be capable of demonstrating proficiency across the following parameters:

1. Multi-Role Frontend Design: Architecture and construct secure, permission-guarded presentation systems serving different user classes.
2. Global State Architecture: Model complex asynchronous caching directories using Redux Toolkit to sync application data states seamlessly.
3. Advanced Form Validation: Implement high-performance data input verification pipelines handling dynamic validations via React Hook Form.

4. Data Visualization Engineering: Translate complex API response arrays into useful interactive data charts using specialized visualization libraries.
5. Component Lifecycle Management: Manage automated network communication hooks efficiently, keeping views updated without causing performance lag[cite: 3].
6. Live Production Infrastructure Deployment: Compile and optimize raw source scripts into secure production-ready application packages hosted on public cloud web servers[cite: 3].